EIN 4360 Practice Exam 3 **Time Limit: 50 minutes**

Instructor: Nicolas Grimaldi
Date: 12/01

QUESTIONS

Write all your answers on a separate

ANSWER SHEET

and SIGN it before starting the exam

INSTRUCTIONS

- The exam is worth a total of **100 points.**
- There are two sections in the exam. Section I has 10 multiple choice questions worth 25 points and section II consists of three (3) descriptive problems worth 75 points.
- Please show all your work clearly. No points will be awarded for anything that is not written in the answer.
- Sharing of calculators is not permitted.
- Please switch off your cell-phone and laptops for the duration of the exam.
- Good luck!

Section I: [25 Points] Multiple Choice Questions.

There are 10 questions in this section each worth **2.5 points**. Circle the letter corresponding to the **BEST** answer on the separate ANSWER SHEET.

(Recommended) Time Limit: **10 minutes**

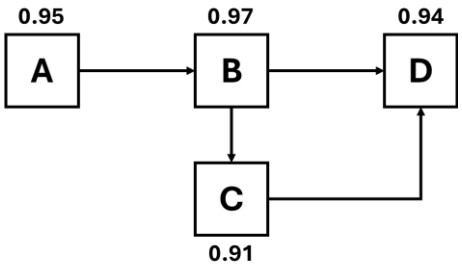
1. (Lecture 11) Which of the following is an example of a fixed cost?
   a. Utilities
   b. Materials
   c. Rent
   d. Shipping
   e. None of the above

2. (Lecture 14) The following is not a pillar of health and safety:
   a. Control
   b. Evaluate
   c. Recognize
   d. Anticipate
   e. These are all pillars of health and safety

3. (Lecture 12) If we want to minimize absolute distance to many facilities, then our goal is to reduce the:
   a. Mean distance
   b. **Median distance**
   c. Variance in distance
   d. Entropy in distance
   e. More than one option is correct

4. (Lecture 13) A dummy activity in project networks denotes a dummy activity, and it's used when:
   a. Defining correct precedence while maintaining unique activity notation
   b. More than two events need to be connected
   c. Allocating additional resources during overlapping activities
   d. Representing activities with zero duration that still consume resources
   e. There are no starting and/or ending events

5. (Lecture 10) Consider the sequence below, where a manufacturing process starts with A and finishes at D. Scrap estimates are denoted next to each process. If B fails, C can be used as a fallback to complete the process. Assume we need to produce 5,000 products. What should our initial quantity be?
   a. 5,702
   b. 5,614
   c. 6,033
   d. 5,421
   e. 4,895



6. (Lecture 13) Which of the following is not a process defined in process management?

a. Executive process
b. Closing process
c. Refining process
d. Controlling process
e. Planning process

7. (Lecture 11) Planning horizon is defined as:

a. The window through which we look in evaluating each investment alternative
b. The future cash flow (e.g., profits and costs) associated with a new investment
c. The time to recover the capital investment you made in an alternative
d. The efficiency of an investment over time
e. None of the above

8. (Lecture 10) An operator will have no idle time when:

a. There are more machines than ideal
b. There are fewer machines than ideal
c. The number of machines is equal to the ideal
d. More than one answer is correct
e. None of the above

9. (Lecture 12) In a Minimax location problem, what happens if we relax the constraint imposed by z? I.e., z is increased from its minimum, optimal value?

a. The optimal location can now be found anywhere between the A-B line
b. The optimal location becomes unsolvable
c. The optimal location can now be found anywhere along a square contour
d. The optimal location can now be found anywhere along two A-B lines
e. None of the above

10. (Lecture 14) Which of the following recommendations might help reduce the reaching hazard shown below?

a. Provide better gripping support
b. Provide equipment to raise and lower items
c. Train workers on team lifting
d. Elevate work areas
e. More than one answer is correct

Section II: [75 Points] Descriptive Problems.

There are three (3) main problems in this section each worth **25 points.**

(Recommended) Time Limit: **40 minutes**

Problem 1 [**25 Points**]
(Lecture 11) Your company wants to purchase a new $300,000 delivery truck to expand its operations. Do not account for salvage cost. The side-by-side cashflow comparisons of buying and not buying the truck are shown below. With this information, determine the:

a) Return on Investment - ROI
b) Total Present Value - TPV
c) Net Present Value - NPV

Year	With Truck	Without Truck
1	$1.4M	$1.2M
2	$1.5M	$1.25M
3	$1.55M	$1.4M
4	$1.6M	$1.45M
5	$1.6M	$1.5M

1)

year	Gain	PV
1	200 k	188.7 k
2	250 k	222.5 k
3	150 k	125.9 k
4	150 k	118.8 k
5	100 k	74.7 k

a) $ROI = \frac{200 + 250 + 150 \times 2 + 100 - 300}{300}$

$ROI = 183\%$

b) $TPV = 188.7k + 222.5k + 125.9k + 118.8k + 74.7k$

$TPV = \$730.6k$

c) $NPV = \$730.6k - \$300k = \$430.6k$

Problem 2 [**25 Points**]
(Lecture 10) To produce one part, our initial material must go through three machines, A, B, and C, in this order. Assume we want to produce 10,000 units. Given the information in the table below, calculate:

a) The initial amount of material required
b) The machine fractions for each machine

	Machine A	**Machine B**	**Machine C**
Standard time (hr)	0.05	0.15	0.10
Scrap estimate (%)	4	7	3
Availability fraction (%)	90	85	90
Historical efficiency (%)	90	95	95
Total availability (hr/month)	100	200	100

2) a) $Q_A = \dfrac{10{,}000}{(1-.03)(1-.07)(1-.04)} = 11{,}547 \leftarrow$ initial amount required = amount required for A

$Q_B = \dfrac{10{,}000}{(1-.07)(1-.03)} = 11{,}085$

$Q_C = \dfrac{10{,}000}{(1-.03)} = 10{,}309$

b) $F_A = \dfrac{11{,}547(.05)}{(0.9)(0.9)(100)} = 7.13$

$F_B = \dfrac{11{,}085(.15)}{(.85)(.95)(200)} = 10.3$

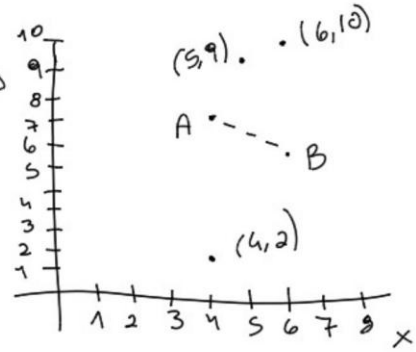
$F_C = \dfrac{10{,}309(.1)}{(.90)(.95)(100)} = 12.1$

Problem 3 [25 Points]

(Lecture 12) Assume we happen to be taking a field trip to the turtle breeders along the east coast of Florida. To ensure they have enough egg-laying locations, Tuck (the turtle from the Wonder Pets) has asked us to build a new location that isn't too far from another egg-laying location. There are currently three egg-laying locations along the shore: (5,9), (4,2), and (6,10).

a) Find the line segment along which a new facility could be placed that minimizes the maximum distance to the other facilities.
b) What is the maximum distance between the new facility you found and any of the other facilities?

3)



$c_1 = 4+2 = 6$

$c_2 = 6+10 = 16$

$c_3 = -4+2 = -2$

$c_4 = -5+9 = 4$

$c_5 = \max(16-6,\ 4-(-2)) = 10$

a) $A = \frac{1}{2}(6-(-2),\ 6-2+10) = \frac{1}{2}(8,\ 14) = (4,7)$

$B = \frac{1}{2}(16-4,\ 16+4-10) = \frac{1}{2}(12,\ 10) = (6,5)$

(solution drawn on coordinate plane)

b) $Z^* = 10/2 = 5$ units

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